Optimization of a Smart Integrated Renewable Energy System for Isolated Rural Villages using Integer Linear Programming

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***Abstract*— Providing energy to areas isolated from the electricity grid through the use of a smart integrated renewable energy system (SIRES) is based on the Agenda 2030 for Sustainable Development, specifically Goal 7, according to the United Nations (UN). This study analyzes the process of identifying the appropriate size of a SIRES considering technical and economic factors (cost of energy-*COE*). An optimization integer linear programming model proposed was used, and a subsequent spatial–temporal analysis of the different variables was conducted. The model comprises locally available renewable energy resources, such as biomass, biogas, wind power, solar photovoltaic, and thermal power. Furthermore, it could be used to determine the energy potential of each of the isolated villages, identifying those areas in which the SIRES could be implemented as a sustainable and economical solution. The design simulates the cost of the initial investment and energy generation in the chosen village. The study also includes the study case in the region of Valparaiso, Chile. Different scenarios were used to optimize the available resources and minimize the *COE* generation. (*Abstract*)**

Keywords*— *smart integrated renewable energy system, optimization model, integer linear programming, off-grid electrification, cost of energy

I. Introduction

The use of non-conventional renewable energy (NCRE) resources such as biomass, biogas, wind, and solar energy in a decentralized mode for the electricity supply has received considerable attention in recent years due to adverse environmental impacts and fuel cost escalation associated with conventional energy generation. These resources have enough potential to become important sources for power generation because of their environmental, social, and economic benefits in addition to public support and government incentives. All this is based on the Agenda 2030 for Sustainable Development, specifically Goal 7, according to the United Nations (UN) [1]. At present, there are approximately 1 billion people functioning without electricity, and 50% of them can be found in Sub-Saharan Africa alone. Fortunately, progress has been made in the past decade regarding the use of renewable electricity from water, solar, and wind power. However, there are several barriers preventing renewable resources from being competitive against conventional energy sources in the current power market, as the supply of energy from these renewable resources cannot be regulated depending on the demand, and integration of these resources with grid network is not easy.

To overcome these problems, the concept of an integrated renewable energy system (IRES) has been propounded which matches the energy requirements of isolated zones far away from the utility grid having locally available renewable energy resources [2]. To keep up with developments in "smart" technologies, IRES was slightly modified and renamed SIRES to reflect the title of smart integrated renewable energy systems, in accordance with Reference [3].

For this system SIRES, some of the resources, such as wind and solar radiation, are highly stochastic and characteristic of a determined site [8]. Others, such as biomass and hydro-energy, are more predictable, since they have a seasonal variation and are characteristic of the site studied [8]. Some loads are more variable than others are, while some can be predicted with greater confidence; then, models or design procedures must consider all these factors and be included in the calculation algorithms.

Different researches have been carried out, directed towards the elaboration of models for designing SIRESs. Examples of these are chronological simulation [10], linear programming [11–12], objective programming [13], and a probabilistic focus that considers the probability of loss of power supply (PPSP) [14]. Chronological simulation requires extensive databases on resources and loads that come from local needs, focusing on linear programming and objective programming and a deterministic use of average seasonal or annual values in their analysis. An interesting study was carried out using a knowledge-based approach; this design improves on previous studies in terms of the efficiency of the algorithm [15].

In other studies, spatial–temporal and spatial analyses have been used to evaluate renewable resources in any location, as [24] shows the wind potential in Malaysia on a nationwide scale. It comprises a spatial–temporal analysis of wind potential and a

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spatial analysis of wind power density in Malaysia. For Chile, a project was carried out to develop a solar explorer [25], allowing the analysis of surface solar radiation. The methodology employed to generate this database was based on the use of a radiative transfer model combined with information inferred from EAST GOES satellite cloudiness and local observations.

As mentioned previously, a Goal of the 2030 Agenda for Sustainable Development presented by the United Nations, as well as a pillar of the Energy Agenda in several country, consists of increasing the use of nonconventional renewable energies for energization of the country and vulnerable areas in terms of access to energy, example [4]. In this sense, main objective at the present study was to find the suitable component sizes and an optimal operation strategy for an isolate village developing and using an optimization model. This model takes into account the necessities and spatiotemporal series of the of isolated villages. The optimization model was applied for the energization off-grid of an isolated villages of the Valparaíso Region in Chile. The results obtained have been used to design and plan an optimal system ensuring reliable and economical power supply to the cluster of villages selected.

II. Methodology

A. Model configuration

Isolated villages are common in developing countries like Chile. Often such villages are off-grid because of the high capital investment in constructing long distribution lines from exiting utility grids. According to the study [5], such villages can be energized by establishing an energy center and installing a distribution line connecting all the houses. Depending upon the local conditions and availability renewable resources its can be integrated with solar energy, wind energy and other possible energy conversion units (e.g. agricultural waste and biomass gasifiers), similar [6]. The proposed optimization model matches the energy requirements with an appropriate energy resource.

The proposed system includes biomass (crop residue and forest foliage), biogas (energy used for electrification, cooking and hot water demand) and solar energy (thermal and photovoltaic) with additional wind energy. The biogas produced from bovine, ovine and porcine animals is first used for cooking and then for electricity generation and hot water demand while solar thermal energy is used for hot water demand and from other renewable energy resources is used for electricity generation only. Additionally, a traditional hot water boiler (THW) was included in the optimization model in order to provide hot water on winter days. The start-up of THW occurs at times that the Solar Thermal Collectors (STC) system can operate offline. The gas supply for the THW system was implemented by means of the biogas plant, with the aim of ensuring the production of sanitary hot water. Fig. 1 shows the line diagram of resources used for electricity generation, hot water production and cooking purposes.

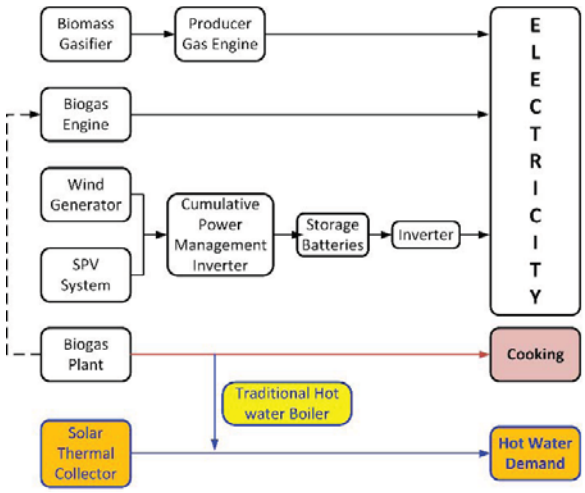

Fig. 1. Configuration of the SIRES optimizacion model.

B. Mathematical model

The proposed optimization model is designed to integrate locally available different renewable energy systems. Each system has its advantages and limitations. The biomass gasifiers are used continuously for short duration while the wind and solar energy are used according to the daily availability or stored for short duration of time. The optimization model can therefore, be planned and designed to overcome their intermittent behaviour of the different energy sources.

In this model, all the energy systems are interdependent on each other and their size varies when small change occurs in the model. The different sets of system sizes are analyzed to get an optimal solution. The developed model is capable to give optimized combination of renewable energy system sizes for a given load profile. The study analyzes the different load profiles and gives different choices to select the best option out of the proposed optimized solutions.

The proposed optimization model has two submodels namely, energy conversion systems and economic submodel. The brief description of each submodel is given below. The detailed mathematical models of biomass, biogas wind and solar systems have already been reported in earlier publications of [6] and the final expressions for the electrical power output of these systems are represented below.

1) Mathematical model for solar photovoltaic system (SPV)

The input energy to SPV system is solar radiation and total solar radiation on a fixed inclined surface is estimated as:

$$I_s = I_b F_b + I_d F_d + (I_b + I_d) F_r \tag{1}$$

where, I_s is total solar radiation on a fixed inclined surface (kW/m^2), I_b and I_d are direct irradiation (or beam) and diffuse solar radiations (kW/m^2), F_b, F_d and F_r are tilt factors for beam, diffuse and reflected part of solar radiations, respectively [7].

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The total estimated radiation, thus, estimated depends on the position of sun in the sky, which varies from month to month. Hourly power output from photovoltaics (PV) system with a surface area A (m^2) on an average day of nth month, when total solar radiation of I_s (kWh/m^2) is incident on PV surface, is given by:

$$P_{PV} = \eta I_s A \tag{2}$$

where, P_{PV} is power generation from PV system (kW), η is system efficiency calculated by standard procedure [8].

2) Mathematical model for wind energy system (WES)

In this model, P_{WT} power generated by wind turbine (kW) at a specific location depends on wind speed at a particular height, power output (P_w), total swept area (A_w), η_w is overall efficiency of gearbox, generator and associated components [9]. In this sense, the power P_{WT} of wind turbine is calculated as follows:

$$P_{WT} = P_w A_w \eta_w \tag{3}$$

3) Mathematical model for for biogas system (BG)

The biogas is assumed to be used mainly to meet the cooking energy and hot water production in isolated villages and thereafter, the surplus is used for the electricity generation [10]. The mathematical model representation is as following:

$$P_{BG} = \eta_b (BG_{total} - BG_{C-H}) \tag{4}$$

where, P_{BG} is power generated by biogas engine generator (kW), η_b is overall efficiency of the biogas fuelled engine and generator system, BG_{total} is total biogas available at that instant (m^3), $BG_{C\text{-}H}$ is biogas consumed for cooking and hot water generation at that instant (m^3).

4) Mathematical model for biomass gasifier system (BM)

Biomass gasifier systems are proven technologies for rural electrification [6], [11]. The biomass gasifier model, mainly, depends on amount of biomass required per hour for the biomass gasifier to produce electricity of rated capacity which is knowed as biomass constant ϕ_B. The mathematical model is given by following equation

$$P_{BM} = \eta_g (\phi_w - \phi_B) \tag{5}$$

where, P_{BM} is power generated by biomass gasifier (kW), η_g is overall efficiency of the gasifier system including efficiency of biomass to producer gas conversion, internal combustion engine and generator and ϕ_w is total biomass available at that instant (kg).

5) Mathematical model for solar thermal collectors system STC

Thermal solar thermal collectors are used to produce sanitary hot water (SHW) consumption of a cluster of villages. A model the size of the STC system was described by [5]. Therefore, the useful power delivered from a concentrator is:

$$P_{STC} = I_b \eta_o A_a - P_{loss} \tag{6}$$

where, P_{STC} is power generated by solar thermal collectors (kW), η_o is collector optical efficiency, A_a is absorber area (m^2) and P_{loss} is power loss in solar collector (kW).

Finally, the total power generation ($P_T(t)$) at any time (t) from optimization model is represented as;

N_i: amount of energy of type i, $i \in \{PV, WT, BM\}$

$$P_T(t) = \sum_{PV=1}^{N_{PV}} P_{PV} + \sum_{WT=1}^{N_{WT}} P_{WT} + P_{BG} + \sum_{BM=1}^{N_{BM}} P_{BM} \tag{7}$$

C. Integer Linear Programming (ILP) optimization model

The ILP problems are optimization problems can be expressed as the minimization or maximization of the linear objective function subject to several constraints over integer variables [12]. We propose an ILP model to obtain the amount of energy that can be produced from each type of resource, minimizing the cost of general production and taking into account the availability of energy and consumption of each energy resource.

1) Parameter

N: amount of houses
D_i: amount of available energy obtained from the type i resource, $\forall i = 1, \dots, 5$
V_{ji}: cost of energy production type j obtained from type i resource, $\forall j = 1, \dots, 3$, $\forall i = 1, \dots, 5$
C_j: house consumption of type j energy, $\forall j = 1, \dots, 3$
R_i: processing performance of resource type i, $\forall i = 1, \dots, 5$
S_i: processor area of type i resource, $\forall i = 1, \dots, 5$

2) Decision variables

COE: cost of energy (calculated in Chilean pesos, app. \$681 CLP ≈ \$1 USD)
P_{ji}: amount of type j energy produced from resource i, $\forall j = 1, \dots, 3$, $\forall i = 1, \dots, 5$
T_i: amount of type i resource processors, $\forall i = 2,3,4$

3) Objective: Minimize

$$Min\ COE = \sum_{j=1}^{3} \sum_{i=1}^{5} V_{ji} P_{ji} \tag{8}$$

4) Constraints

$$\sum_{i=1}^{5} P_{ji} \geq C_j * N, \quad \forall j = 1, \dots, 3 \tag{9}$$

$$\sum_{j=1}^{3} P_{ji} \Big/ R_i \leq D_j, \quad i \in \{1,5\} \tag{10}$$

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$$\sum_{j=1}^{3} P_{ji} \Big/ R_i \leq D_j T_i, \;\; i \in \{2,3,4\} \tag{11}$$

$$P_{ji} \geq 0 \tag{12}$$

$$T_i \in \mathbb{N} \tag{13}$$

Equation 10 and 11 take into account the energy viability. Renewable energy resources must exist, which have an energy potential high enough to cover the expected demand of the user, considering the performance of the installation and as far as possible not having energy deficit at any time of the year [13]–[15].

The ILP model was developed in MINIZINC software version 2.1.7 to optimization problem [16]. The MINIZINC is an open-source constraint solving language designed to addressing optimization problems [17]. Several research in this field have solved this type of models using this solver [18], [19].

III. CASE STUDY

A. Study area

It was estimated that in Chile there are more than 40.000 homes without access to electricity, equivalent to 5% of the rural population [20]. Specifically, in the region of Valparaiso by means of the Energy Regional Ministry of Energy Secretariat (SEREMI) was reported and located the rural villages off-the-grid electrical. Approximately, 165 rural homes in the region are isolated of the grid.

The study of [5] shown the resources, load and demand assessment across different isolated villages in the region of Valparaiso. The potentials were based on the most common sources of NCRE present in the region, disregarding less developed technology that implies a much higher investment cost (such as geothermal, tidal, and wave energy). Comparing sectors, it can be seen that the highest average energy contribution is Cabildo. This village is composing by 32 houses. Fig. 2 shows the geographic location of the study area.

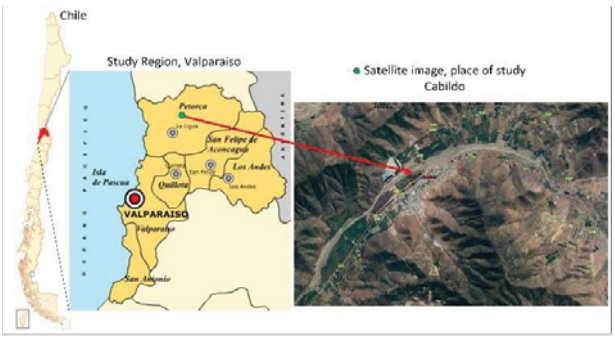

Fig. 2. Location of the study area.

B. Resources and load/demand assessment

1) Resource assessment

Aforementioned, in order to assess the potential of NCRE, an extensive survey was conducted and data about the availability of intensity of solar radiation, wind speed and biomass was collected by [5]. The potential assessment has been carried out as per the methodology adopted in the earlier study carried out by the authors [21]. The data indicates that biogas resource is the maximum followed by solar energy, biomass including crop residues and forest foliage and wind potential. Table I shows the annual available resources on the scenario proposed. These different scenarios are considered to optimize the available resources and minimize the *COE* generation that it is the economic constraint for implementation of these systems. A brief description of the scenarios follow.

a) Scenario 1. Critical month

In the first scenario, NCRE systems are considered according to the critical month [22], that is, the condition where the solar and wind energy has the minimum availability. These renewable sources depend highly on local climate conditions or season of the year. The electricity is generated by mean WSE, biomass (crop residue and forest foliage) and biogas are considered along with SPV system only. The STC and biogas are used to heat water. Finally, for cooking only the biogas is employed.

b) Scenario 2. Critical month

The biomass and biogas potential has been negligible because in the village selected could be impossible to estimate the availability under certain conditions.

c) Scenario 3. Anual average

The last scenario is similar to scenario 1, but the renewable resources are calculated from annual availability.

2) Load/demand assessment

The energy consumption of the village selected is naturally dependent the types of energy used in each home: electricity consumption for electrical appliances and other electrical items and thermal consumption, mainly used for cooking and SHW. Regarding, the energy contribution required of a SIRES needs to be known to the highest degree of accuracy possible to avoid an energy deficit or inadequate design of the installation.

The proposed case is of a single-family home comprising four people. In [5] was calculated the energy demand for a home isolated from the grid of the types “basic” and “equipped”, defined by [23]–[25]. The annual energy demand is 1328.6 [kWh/yr], 2054.95 [kWh/yr] and 766.5 [kWh/yr] for electricity, sanitary hot water and cooking, respectively.

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TABLE I. ESTIMATED RENEWABLE ENERGY RESOURCES AND COE IN THE STUDY AREA

Scenario	Available Resources				
	SPV ($kWh/m^2/yr$)	*WES* ($kWh/m^2/yr$)	*BG* (kWh/yr)	*BM* (kWh/yr)	*STC* (kWh/yr)
I	1109.6	158.05	1558.6	912.5	1109.6
II	1109.6	158.05	0	0	1109.6
III	1879.8	259.2	1558.6	912.5	1879.8
COE ($/kWh)	**183**	**963**	**90 for electricity 42 for hot water 43 for cooking**	**82**	**92**

IV. RESULTS AND DISCUSSION

As discussed above, the objective of this study is to develop an optimal operation strategy for a village off-the-grid using and optimization model. The developed model has been optimized for different scenarios, described above, using MINIZINC. The optimization model determines the size and technical–economic viability for the feasibility of implementing an SIRES, in order to satisfy the energy demands (electrical and heat energy) in a specific isolated village, Cabildo. The system sizes for the different sources of renewable energy have been considered in accordance with the manufacturer's specifications, and the four seasons were taken in account, and the resulting required loads were calculated.

The percentage contribution of each renewable energy resources for the different scenarios are given in Table II, which shows that implementation of resources even in small magnitude, reduces the *COE*.

TABLE II. COE FOR ALL SCENARIOS AND PERCENTAGE CONTRIBUTION RENEWABLE ENERGY RESOURCES

Scenario	Energy Production	Available Resources (%)					COE ($/kWh)
		SPV	*WES*	*BG*	*BM*	*STC*	
I	Electricity	6.2	0	35.4	58.4	-	**81.6**
	Hot water	-	-	4.3	-	95.7	
	Cooking	-	-	100	-	-	
II	Electricity	100	-	-	-	-	**138.9**
	Hot water	-	-	-	-	100	
	Cooking	-	-	-	-	-	
III	Electricity	-	-	41.6	58.4	-	**80.8**
	Hot water	-	-	0.26	-	99.74	
	Cooking	-	-	100	-	-	

From the previous table, it is observed that in scenario I, the energies resources used correspond to biomass, biogas and solar thermal with the lowest *COE*. When consuming them completely it is necessary to use part of the generated energy by solar energy. The model minimizes the use of solar energy and cancelling the use of the wind that has the highest *COE*.

In scenario II, assuming a negligible consumption of biomass and biogas, the needs of SHW are satisfied by means STC. In this case is required a STC capture area greater than that of scenario I. In addition, the scenario II has the highest *COE*. Regarding, the scenario III has a behaviour similar to scenario I, where the main difference is availability of solar and wind energy, representing a difference of 40.9% and 39%, respectively. In this case, the *COE* is only 1% greater than scenario I because the production of electricity is only done through biomass gasifier and biogas which are the resources that have the lowest *COE*. In addition, for this scenario, almost all the need for hot water is done through STC. This implies that the energy biomass a biogas are the main resources that reduces the *COE* generation drastically and the scenario III is found to be the best among the other three scenarios considered on the basis of *COE* constraint and energy viability.

Finally, it should be noted that in all scenarios, the wind energy not is used on the energy generation due to that has the highest *COE*, and yet there is no area constraint for solar energy. In addition, whenever it is necessary to produce electricity and there is no biogas or biomass disposal, it is more profitable to occupy solar energy.

V. CONCLUSIONS

In view of the worldwide energy shortages and environmental impacts of fossil fuel used, the concept of SIRES has been considered for the Chilean off-the-grid village for the meeting their energy needs. The unit cost of energy from each resource has been calculated on the basis of available resources and the demand of the village selected. An SIRES model consisting of PV, WES, BM, BG and STC has been constructed and optimized using MINIZINC software version 2.1.7. the study was conducted for three scenarios selected according to the energy availability. The results indicate that model can provide feasible solution in order to minimize the *COE*. The results further indicate that for the study village, either an SIRES consisting of the biomass gasifier engine system, biogas and solar thermal energy can provide a feasible solution in terms of energy fulfillments (electricity, hot water and cooking).

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REFERENCES

[1] 'Goal 7: Sustainable Development Knowledge Platform'. [Online]. Available: https://sustainabledevelopment.un.org/sdg7. [Accessed: 24-May-2019].

[2] R. Ramakumar, 'Role of renewable energy in the development and electrification of remote and rural areas', in *IEEE Power Engineering Society General Meeting, 2004.*, Denver, CO, USA, 2004, vol. 2, pp. 2104–2106.

[3] Z. Maheshwari and R. Ramakumar, 'Smart Integrated Renewable Energy Systems (SIRES): A Novel Approach for Sustainable Development', *Energies*, vol. 10, no. 8, pp. 1145–22, Aug. 2017.

[4] A. Rajeev and K. Shanmukha Sundar, 'Design of an off-grid PV system for the rural community', in *2013 International Conference on Emerging*

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Trends in Communication, Control, Signal Processing and Computing Applications (C2SPCA), Bangalore, India, 2013, pp. 1–6.

[5] Y. Masip, A. Gutierrez, J. Morales, A. Campo, and M. Valín, 'Integrated Renewable Energy System based on IREOM Model and Spatial–Temporal Series for Isolated Rural Areas in the Region of Valparaiso, Chile', *Energies*, vol. 12, no. 6, pp. 1110–19, 2019.

[6] A. B. Kanase-Patil, R. P. Saini, and M. P. Sharma, 'Sizing of integrated renewable energy system based on load profiles and reliability index for the state of Uttarakhand in India', *Renewable Energy*, vol. 36, no. 11, pp. 2809–2821, Nov. 2011.

[7] J. A. Duffie and W. A. Beckman, *Solar engineering of thermal processes*, 4. ed. New York: Wiley, 2013.

[8] M. Kolhe, K. Agbossou, J. Hamelin, and T. K. Bose, 'Analytical model for predicting the performance of photovoltaic array coupled with a wind turbine in a stand-alone renewable energy system based on hydrogen', *Renewable Energy*, vol. 28, no. 5, pp. 727–742, Apr. 2003.

[9] R. Chedid, H. Akiki, and S. Rahman, 'A decision support technique for the design of hybrid solar-wind power systems', *IEEE Trans. On energy Conversion*, vol. 13, no. 1, pp. 76–83, Mar. 1998.

[10] A. Mudhoo, Ed., *Biogas production: pretreatment in anaerobic digestion*. Beverly, MA: Scrivener, 2012.

[11] A. K. Akella, M. P. Sharma, and R. P. Saini, 'Optimum utilization of renewable energy sources in a remote area', *Renewable and Sustainable Energy Reviews*, vol. 11, no. 5, pp. 894–908, Jun. 2007.

[12] A. Schrijver, *Theory of linear and integer programming*, Reprinted. Chichester: Wiley, 2000.

[13] L. C. G. Valente and S. C. A. de Almeida, 'Economic analysis of a diesel/photovoltaic hybrid system for decentralized power generation in northern Brazil', *Energy*, vol. 23, no. 4, pp. 317–323, Apr. 1998.

[14] E. A. Alsema and E. Nieuwlaar, 'Energy viability of photovoltaic systems', *Energy Policy*, vol. 28, no. 14, pp. 999–1010, Nov. 2000.

[15] M. K. Deshmukh and S. S. Deshmukh, 'Modeling of hybrid renewable energy systems', *Renewable and Sustainable Energy Reviews*, vol. 12, no. 1, pp. 235–249, Jan. 2008.

[16] 'MiniZinc'. [Online]. Available: https://www.minizinc.org/. [Accessed: 30-May-2019].

[17] N. Nethercote, P. J. Stuckey, R. Becket, S. Brand, G. J. Duck, and G. Tack, 'MiniZinc: Towards a standard CP modelling language', in *Principles and Practice of Constraint Programming – CP 2007*, vol. 4741, C. Bessière, Ed. Berlin, Heidelberg: Springer Berlin Heidelberg, 2007, pp. 529–543.

[18] J. Boland, L. R. Cirocco, M. Belusko, F. Bruno, and P. Pudney, 'Controlling stored energy in a concentrating solar thermal power plant to maximise revenue', *IET Renewable Power Generation*, vol. 9, no. 4, pp. 379–388, May 2015.

[19] S. He, A. Liebman, A. Rendl, M. Wallace, and C. Wilson, 'Modelling RTP-based residential load scheduling for demand response in smart grids', 2014.

[20] 'Energías renovables ayudarían al desarrollo de zonas rurales y extremas de Chile'. [Online]. Available: http://www.reportesostenible.cl/Energias-renovables-ayudarian-al-desarrollo-de-zonas-rurales-y-extremas-de-Chile. [Accessed: 28-May-2019].

[21] A. B. Kanase-Patil, R. P. Saini, and M. P. Sharma, 'Integrated renewable energy systems for off grid rural electrification of remote area', *Renewable Energy*, vol. 35, no. 6, pp. 1342–1349, Jun. 2010.

[22] R. Ramakumar, 'Technical and socio-economic aspects of solar energy and rural development in developing countries', *Solar Energy*, vol. 19, no. 6, pp. 643–649, 1977.

[23] Ministerio de Energía, 'NCh2911/2-Energía de sistemas híbridos-especificaciones para el uso de energías renovables en electrificación rural descentralizada—Parte 2: Directrices para el diseño y funcionamiento del sistema'. Gobierno de Chile: Santiago, Chile, 2010.

[24] Cámara Chilena de la Construcción, 'Estudio de Usos Finales y Curva de Oferta de la Conservación de la Energía en el Sector Residencial.' Gobierno de Chile: Santiago, Chile, 2010.

[25] Ministerio de Economía Fomento y Reconstrucción, 'Norma técnica que determina algoritmo para la verificación de la contribución solar mínima de los sistemas solares térmicos acogidos a la franquicia tributaria de la LEY N° 20.897'. Gobierno de Chile: Santiago, Chile, 2016.

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